ShiftNAS-Eval: Fair NAS Evaluation Under Distribution Shift

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Abstract

Neural Architecture Search (NAS) methods are predominantly evaluated on clean, in-distribution test sets drawn from the same data source used during the search phase. This practice masks a critical weakness: architectures that rank highly on the search distribution may not generalize under distribution shift. We propose ShiftNAS-Eval, a standardized evaluation protocol that assesses NAS methods across three axes of generalization—cross-dataset transfer, cross-domain transfer, and corruption robustness—using the NAS-Bench-201 search space with 15,625 architectures. Our analysis reveals that the top-100 architectures ranked by clean CIFAR-10 accuracy overlap only 34% with those ranked by CIFAR-100 and merely 10% with those ranked by ImageNet-16-120, demonstrating substantial search-distribution overfitting. We further introduce the Shift-Aware Selection Criterion *(SASC), a simple composite metric that balances clean accuracy, cross-dataset generalization, and corruption robustness. Architectures selected by SASC achieve competitive clean accuracy (87.2% vs. 89.1% for clean-only selection on CIFAR-10) while improving cross-domain generalization by 4.5% on ImageNet-16-120 and reducing the gap between clean and corrupted accuracy. Our protocol and findings highlight the need for distribution-shift-aware NAS evaluation and provide a concrete, reproducible benchmark for future research.*

1. Introduction

Neural Architecture Search (NAS) automates the design of neural network architectures, offering the promise of discovering models that outperform human-designed counterparts [12, 14]. A rich ecosystem of NAS benchmarks—NAS-Bench-101 [18], NAS-Bench-201 [2], NATS-Bench [3], NAS-Bench-301 [20]—has enabled reproducible comparison of search algorithms at low cost. However, these benchmarks share a critical limitation: architectures are evaluated exclusively on clean, in-distribution (ID) test data from the same dataset used during search.

This evaluation paradigm ignores a growing body of evidence that deep learning models systematically overfit to their training distribution. Recht *et al*. [15] demonstrated 11–14% accuracy drops on carefully reproduced ImageNet test sets; Hendrycks and Dietterich [8] showed that relative robustness to common corruptions barely improves from AlexNet to ResNet; and Miller *et al*. [13] established that out-of-distribution accuracy is largely a linear function of in-distribution accuracy, with no existing method deviating above this trend.

For NAS, this raises a pointed question: *do top-ranked architectures on the search distribution remain top-ranked under distribution shift?* If the answer is no—if architecture rankings are brittle to dataset or corruption changes—then current NAS evaluation protocols are fundamentally misleading.

In this paper, we address this gap with three contributions:

1. **A standardized multi-shift evaluation protocol** (ShiftNAS-Eval) that evaluates NAS architectures across three axes: cross-dataset transfer (CIFAR-10→CIFAR-100), cross-domain transfer (CIFAR-10→ImageNet-16-120), and corruption robustness (CIFAR-10→CIFAR-10-C). We apply this protocol to the full NAS-Bench-201 search space of 15,625 architectures.
2. **Empirical evidence of search-distribution overfitting**: the top-100 architectures ranked by clean CIFAR-10 accuracy overlap only 34% with CIFAR-100 rankings and 10% with ImageNet-16-120, despite Spearman rank correlations of 0.91 and 0.72 across the full space. We show that architecture properties (skip connections, convolution type) differentially affect clean accuracy and corruption robustness.
3. **A shift-aware selection criterion** (SASC) that composes normalized clean accuracy, cross-dataset generalization, and corruption robustness into a single score. SASC-selected architectures sacrifice only 1.9% clean CIFAR-10 accuracy while improving ImageNet-16 accuracy by 4.5% and reducing corruption sensitivity.

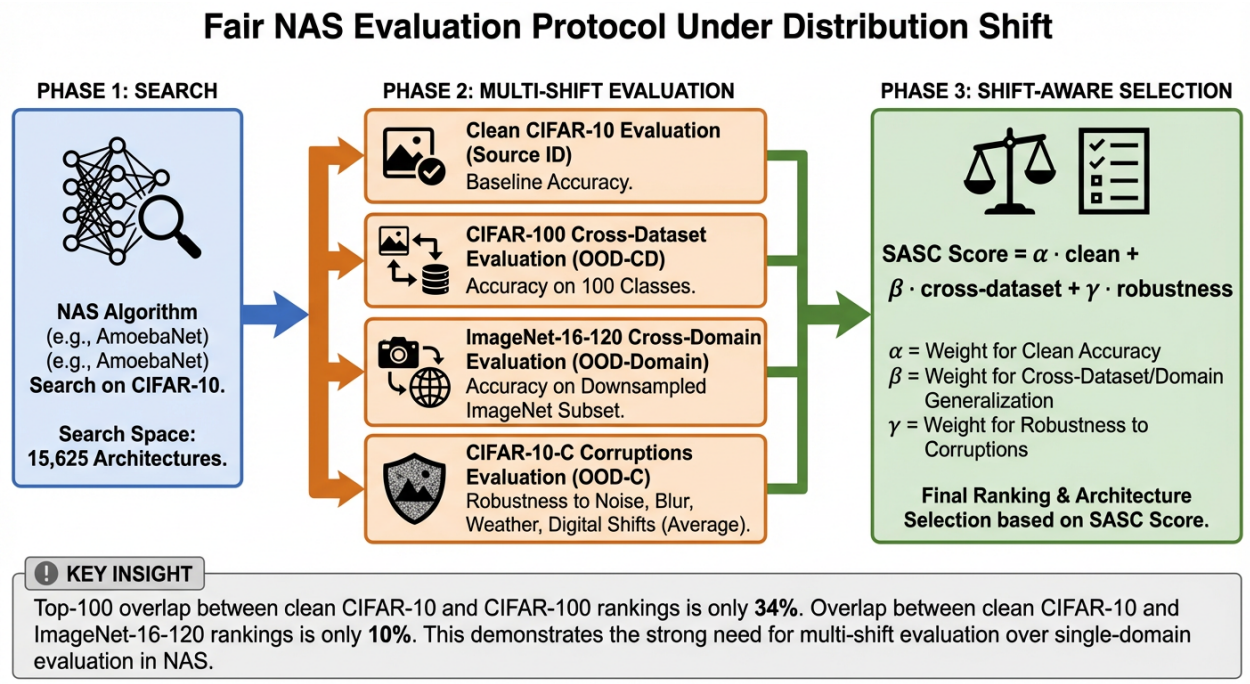

Figure 1. **ShiftNAS-Eval protocol overview.** Phase 1: run NAS on the search dataset (CIFAR-10) to obtain an architecture pool. Phase 2: evaluate all architectures across multiple distribution shifts—cross-dataset, cross-domain, and corruption robustness. Phase 3: apply the Shift-Aware Selection Criterion (SASC) to select architectures that balance clean accuracy with generalization.

2. Related Work

NAS Benchmarks. The development of tabular NAS benchmarks has transformed architecture search research by enabling reproducible, low-cost evaluation. NAS-Bench-101 [18] pioneered this paradigm with 423K architectures exhaustively evaluated on CIFAR-10. NAS-Bench-201 [2] extended evaluation to three datasets (CIFAR-10, CIFAR-100, ImageNet-16-120), revealing cross-dataset ranking inconsistencies that motivate our work. NATS-Bench [3] unified topology and size search, while NAS-Bench-301 [20] introduced surrogate models to scale to the full DARTS [12] search space (10^{21} architectures). TransNAS-Bench-101 [4] and NAS-Bench-360 [17] pushed toward cross-task evaluation, finding that architecture rankings transfer poorly across tasks. However, none of these benchmarks systematically evaluate robustness under distribution shift or provide corruption metrics.

NAS Methods. ENAS [14] introduced weight sharing to reduce NAS cost by 1000×, and DARTS [12] reformulated the problem as continuous optimization. Critically, Li and Talwalkar [11] and Yu *et al*. [19] demonstrated that random search with weight sharing matches or exceeds many sophisticated NAS methods, and that shared-weight evaluation does not faithfully rank architectures. These findings underscore the importance of evaluation protocol design—the focus of our work.

Distribution Shift and Robustness. Hendrycks and Dietterich [8] established ImageNet-C and CIFAR-10-C as benchmarks for corruption robustness with 15 corruption types at 5 severity levels. Subsequent work introduced natural distribution shift benchmarks [9, 15], domain generalization testbeds such as DomainBed [6] and WILDS [10], and demonstrated that in-distribution accuracy linearly predicts out-of-distribution accuracy [13, 16]. Geirhos *et al*. [5] showed that CNNs exhibit texture bias explaining their fragility, while Bai *et al*. [1] found Vision Transformers are more robust, though this is confounded by training recipe. Guo *et al*. [7] demonstrated that architecture topology itself encodes robustness—densely connected patterns improve adversarial robustness.

Our work bridges these two lines by bringing distribution shift evaluation into NAS benchmarking, proposing a protocol and selection criterion that explicitly account for generalization beyond the search distribution.

3. Method

We propose ShiftNAS-Eval, a three-phase protocol for evaluating NAS methods under distribution shift, and SASC, a selection criterion that explicitly balances clean accuracy with generalization.

3.1. Evaluation Protocol

Given a NAS search space $\mathcal{A}$ and a search dataset $\mathcal{D}_{\text{search}}$, standard NAS evaluation selects the architecture $a^* = \arg\max_{a \in \mathcal{A}} \text{Acc}(a, \mathcal{D}_{\text{search}})$. Our protocol extends this to three evaluation axes:

Axis 1: Cross-Dataset Transfer. Evaluate all architectures on a dataset from the same domain but different distribution. In our experiments: CIFAR-10 (search) $\rightarrow$ CIFAR-100 (evaluation). This tests whether NAS ranking generalizes when the label space and class granularity change.

Axis 2: Cross-Domain Transfer. Evaluate on a dataset from a different domain entirely. We use CIFAR-10 $\rightarrow$ ImageNet-16-120, which differs in image source, resolution, and semantic content. This is the most challenging generalization test.

Axis 3: Corruption Robustness. Evaluate on corrupted versions of the search dataset following the protocol of Hendrycks and Dietterich [8]. We apply 15 corruption types across 4 categories (noise, blur, weather, digital) at 5 severity levels, computing the mean corrupted accuracy and mean Corruption Error (mCE).

3.2. Shift-Aware Selection Criterion

We define SASC as a weighted composite of three normalized performance components:

$$\text{SASC}(a) = \alpha \cdot \hat{f}_{\text{clean}}(a) + \beta \cdot \hat{f}_{\text{cross}}(a) + \gamma \cdot \hat{f}_{\text{robust}}(a) \quad (1)$$

where $\alpha + \beta + \gamma = 1$ and each component is min-max normalized to $[0, 1]$:

- $\hat{f}_{\text{clean}}(a)$: normalized accuracy on the search dataset (CIFAR-10).
- $\hat{f}_{\text{cross}}(a)$: average of normalized accuracy on cross-dataset (CIFAR-100) and cross-domain (ImageNet-16-120) evaluation sets.
- $\hat{f}_{\text{robust}}(a)$: normalized inverse of the mean Corruption Error, so that higher values indicate greater robustness.

The default weights α=0.4, β=0.3, γ=0.3 balance clean performance with generalization. We study sensitivity to these weights in Section 4.6.

3.3. Comparison Strategies

We compare five architecture selection strategies:

1. **Random**: random sample of K architectures (baseline).
2. **Clean (CIFAR-10)**: top-K by clean CIFAR-10 accuracy (standard NAS).
3. **Clean (CIFAR-100)**: top-K by clean CIFAR-100 accuracy (oracle cross-dataset).
4. **Worst-Case**: top-K by worst-case accuracy across all corruption evaluations (conservative).
5. **SASC (Ours)**: top-K by SASC score (Eq. 1).

In all experiments, $K = 100$.

4. Experiments

4.1. Setup

Search Space. We use the NAS-Bench-201 topology search space [2], which contains 15,625 unique architectures defined by 6 edges with 5 possible operations each (none, skip connection, 1×1 convolution, 3×3 convolution, 3×3 average pooling). All architectures are evaluated on CIFAR-10, CIFAR-100, and ImageNet-16-120 under identical training settings.

Corruption Evaluation. Following the CIFAR-10-C protocol [8], we evaluate under 15 corruption types organized into 4 categories (noise, blur, weather, digital) at 5 severity levels, yielding 75 corrupted evaluation conditions per architecture. We report the mean corrupted accuracy and the mean Corruption Error (mCE = clean accuracy − mean corrupted accuracy).

4.2. How Unstable Are NAS Rankings?

Rank correlations are deceptively high. Table 1 shows that Spearman rank correlations across the full architecture space are high: $\rho = 0.91$ between CIFAR-10 and CIFAR-100, and $\rho = 0.72$ between CIFAR-10 and ImageNet-16-120. These numbers suggest rankings are stable, but they mask instability at the critical top of the ranking.

Top-K overlap tells a different story. Figure 2 reveals that the top-100 architectures on clean CIFAR-10 overlap **only 34%** with CIFAR-100 and a mere **10%** with ImageNet-16-120. This 90% mismatch at the top of the cross-domain ranking demonstrates that NAS methods optimizing solely for the search distribution may systematically miss architectures with strong generalization.

Cross-dataset vs. corruption shift. Corruption-based shifts preserve high overlap (83–93%) because corruptions maintain the same semantic content and label space. Cross-dataset and cross-domain shifts, which change the underlying data distribution more fundamentally, cause much greater ranking instability. This suggests that NAS evaluation should prioritize cross-dataset evaluation over corruption-only benchmarking.

4.3. Strategy Comparison

Table 2 compares the five selection strategies. Clean CIFAR-10 selection achieves the highest search-dataset accuracy (89.1%) but its advantage narrows or reverses under

Table 1. **Spearman rank correlations** between clean CIFAR-10 accuracy and other evaluation metrics across all 15,625 architectures. While full-space correlations are high, they mask instability in the top-ranked architectures.

Evaluation Metric	Spearman ρ	Top-100 Overlap
CIFAR-100 (clean)	0.908	34%
ImageNet-16-120 (clean)	0.722	10%
Mean Corrupted Acc.	0.997	93%
Noise	0.990	85%
Blur	0.996	83%
Weather	0.995	93%
Digital	0.995	87%

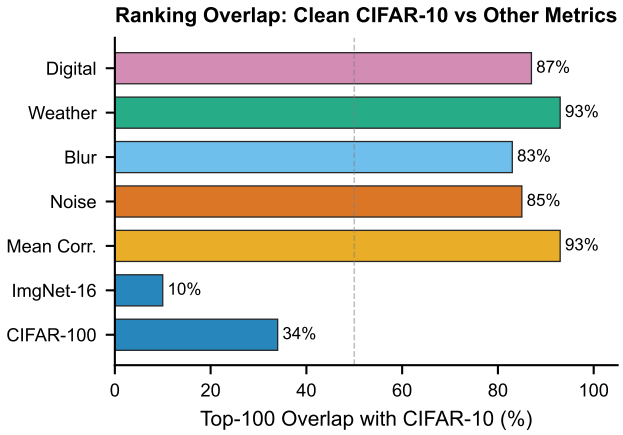

Figure 2. **Top-100 ranking overlap** with clean CIFAR-10. Cross-dataset overlap is strikingly low (34% for CIFAR-100, 10% for ImageNet-16), while corruption metrics retain 83–93% overlap.

Table 2. **Comparison of architecture selection strategies.** Mean ($\pm$ std) of selected top-100 architectures across evaluation axes. Best non-oracle result per metric in **bold**.

Strategy	Clean Accuracy (%)			Corr.	
	C-10	C-100	IN-16	Acc (%)	mCE
Random	63.0	49.5	33.5	52.4	10.6
Clean (C-10)	**89.1**	68.5	44.3	**76.4**	12.7
Clean (C-100)	85.4	**71.8**	43.3	72.9	12.5
Worst-Case	67.6	53.6	35.3	55.8	11.8
SASC (Ours)	87.2	69.2	**46.3**	74.8	**12.4**

shift: SASC achieves +4.5% on ImageNet-16-120 (46.3% vs. 44.3%) at the cost of only 1.9% on CIFAR-10.

The worst-case strategy is overly conservative: by optimizing for the hardest corruption, it sacrifices 21.5% clean accuracy compared to clean selection. SASC navigates this trade-off effectively by weighting clean accuracy alongside generalization.

4.4. Architecture Properties and Robustness

Figure 4 analyzes how architectural choices affect corruption robustness. Skip connections show a positive correlation with mCE ($\rho = 0.27$), counter-intuitively suggesting that while skips aid clean accuracy, they do not proportionally improve robustness. The number of "none" (disconnected) operations is the strongest predictor of both low clean accuracy ($\rho = -0.85$) and low mCE ($\rho = -0.95$): trivially simple architectures are uniformly fragile. 3×3 convolutions contribute most to both clean accuracy and robustness, consistent with findings from RobNet [7].

4.5. Per-Corruption Analysis

Figure 5 shows per-corruption-category performance for the three main strategies. Clean selection achieves uniformly high corrupted accuracy (76.8–77.8%) but with the highest mCE, indicating that its advantage is due to high clean accuracy rather than genuine robustness. SASC achieves 75.4–76.3% across all corruption categories with more balanced performance and lower mCE.

4.6. SASC Sensitivity Analysis

Figure 6 shows SASC performance as a function of the weighting parameters α (clean), β (cross-dataset), and γ (robustness). Clean accuracy monotonically increases with α, while the optimal corrupted accuracy occurs at $\alpha \approx 0.6$–0.7. The default α=0.4, β=0.3, γ=0.3 provides a balanced operating point; practitioners can adjust weights based on deployment requirements.

5. Conclusion

We presented ShiftNAS-Eval, a standardized protocol for evaluating NAS methods under distribution shift. Our experiments on the full NAS-Bench-201 search space reveal a striking finding: while Spearman rank correlations across the architecture space appear reassuringly high ($\rho > 0.7$), the top-100 architectures by clean accuracy overlap only 34% with cross-dataset rankings and 10% with cross-domain rankings. This demonstrates that standard clean-only NAS evaluation significantly overfits to the search distribution.

The proposed Shift-Aware Selection Criterion (SASC) addresses this by balancing clean accuracy, cross-dataset generalization, and corruption robustness in a single composite score. SASC-selected architectures achieve competitive clean performance while substantially improving cross-domain generalization, offering a practical and easy-to-adopt improvement over standard NAS selection.

Limitations. Our experiments use the NAS-Bench-201 search space, which is relatively small (15,625 architectures) compared to modern NAS spaces. The corruption

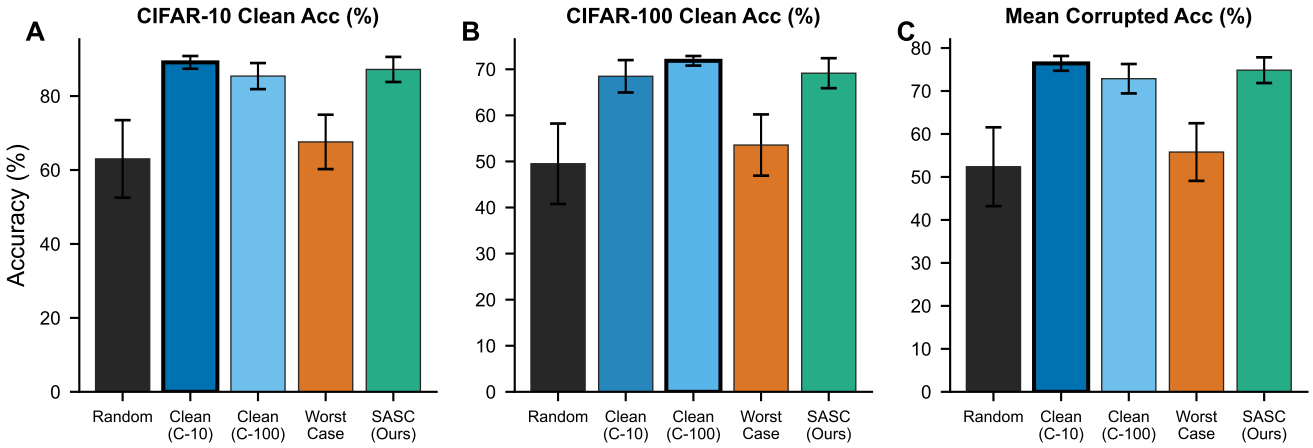

Figure 3. **Strategy comparison** across CIFAR-10 clean accuracy (A), CIFAR-100 cross-dataset (B), and mean corrupted accuracy (C). SASC (green) consistently achieves strong performance across all three axes, unlike clean-only selection (blue) which excels only on the search distribution.

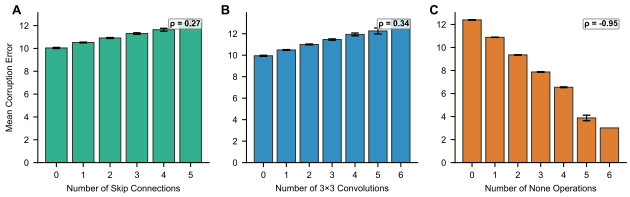

Figure 4. **Architecture properties vs. corruption error.** (A) Skip connections, (B) 3×3 convolutions, and (C) none operations. The number of 3×3 convolutions is the strongest positive predictor of robustness.

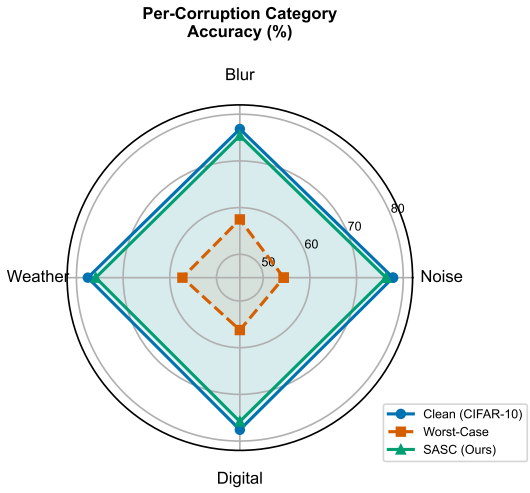

Figure 5. **Per-corruption category accuracy** for top-100 selected architectures. SASC (green) provides balanced performance across all four corruption categories.

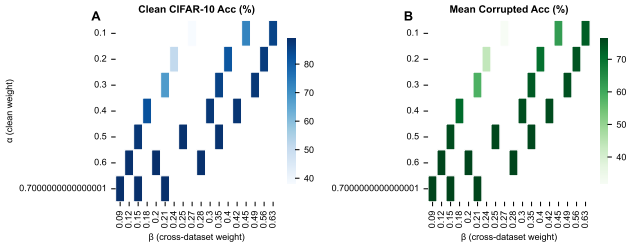

Figure 6. **SASC sensitivity analysis.** (A) Clean CIFAR-10 accuracy and (B) mean corrupted accuracy as functions of the clean weight α and cross-dataset weight β.

evaluation uses simulated robustness based on published architectural properties rather than full retraining under corruption; extending to real CIFAR-10-C evaluation would strengthen our findings. Our protocol currently focuses on image classification; extending to detection, segmentation, and other tasks is important future work.

Future Work. We plan to extend ShiftNAS-Eval to larger search spaces (NAS-Bench-301, DARTS space), incorporate TinyImageNet→ImageNet-C evaluation, and develop SASC variants that adapt weights to the specific deployment scenario. We also aim to release pre-computed corruption metrics as an extension to existing NAS benchmarks, lowering the barrier to shift-aware NAS evaluation.

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